

Electrical conductance in a porous medium

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ABSTRACT

The Hanai-Bruggeman equation is applied to a model of a conductive rock matrix immersed in water. The result is a general two-component equation which relates the electrical properties of a formation to the electrical properties of the rock matrix and the water at any frequency. The low-frequency limit is compared with previous theoretical and experimental work on sand formations. From a comparison of the three-resistor model of Wyllie and Southwick, the Waxman-Smits model, the dual water model of Clavier et al (1977) and the proposed model, it is found that the only representation which is viable, conceptually sound, and in agreement with experiment, is the last one.

INTRODUCTION

The interpretation of electrical conductivity measurements in porous material has applications in many areas of science and technology. In particular, it is of considerable importance to the oil industry for distinguishing hydrocarbons from water in formation pores. Although this paper has applications in other areas, it is primarily concerned with the interpretation of electrical well logging measurements.

The basic relationship for interpreting the resistivity log is Archie's law which was formulated in 1942. However, since its application is limited to "clean" sands, i.e., formation matrices which are poor conductors, it has been modified and expanded to "explain" resistivity measurements over a wide range of rock formations. In 1954, Wyllie and Southwick suggested that an aggregate of conductive particles saturated with a conducting electrolyte could be modeled with a three resistor network. In 1957, de Witte discovered an empirical relationship which is a special case of the Wyllie and Southwick model but is more general than Archie's law. In 1968, Waxman and Smits proposed the first extensive model of shaly sand formations. They proposed that clay particles contribute exchange cations to the electrolyte thereby increasing the conductivity of the formation. In 1977 Clavier et al suggested a model based on the assumption that the exchange cations contribute to the conductivity of a clay water which is spatially separated from the bulk water. Consequently, it is called the dual water model. In all this work Archie's empirical law is assumed.

Recently, however, Sen (1980) showed how Archie's law follows from the work of Bruggeman (1935) and Hanai (1960, 1961) assuming that nonconducting rock particles are dispersed in a continuous phase of saline water. In the present work this approach is reinterpreted and expanded to include a conducting lattice-like matrix.

This paper is divided into three major parts. The first is a critical review of the basic assumptions which form the foundation of the earlier models. In the second part a model, which is based on the work of Bruggeman and Hanai, is formulated. The model is of a general nature which relates the electrical properties of any heterogeneous two-component mixture to the properties of the individual components at any frequency. However, only the low-frequency limit is considered in detail. In the final section, this low-frequency case is applied to shaly sand formations and compared with previous work.

PREVIOUS THEORETICAL MODELS OF SHALY SAND FORMATIONS

The three-resistor model

One of the first attempts to understand the electrical conductivity of a water-saturated shaly sand formation is due to Wyllie and Southwick (1954). Their model is based on the assumption that the conductivity of an aggregate of conductive particles saturated with a conducting electrolyte can be represented by three resistive elements in parallel (Figure 1c). They are (1) an element consisting of particles in electrical contact with each other, (2) an element consisting of the electrolyte, and (3) an element consisting of particles and electrolyte in series with each other. Mathematically one can then write the conductance of a formation as

$$\sigma_0 = \frac{\sigma_r \sigma_w}{W \sigma_r + Z \sigma_w} + X \sigma_r + Y \sigma_w, \quad (1)$$

where σ_0 = conductivity of formation, σ_r = conductivity of rock matrix, σ_w = conductivity of electrolyte, and W , X , Y , Z = geometrical factors.

Sauer et al (1955) performed an experiment with cation exchange resin spheres immersed in four different electrolytic solutions. Such a system is electrically similar to shaly sand formations. They analyzed their results in terms of equation (1). They found a good fit to their data and therefore concluded that the curvature in Figure 2 could be explained by the geome-

try of the system. That is, the curvature at small σ_w can be explained by the first term of equation (1) which describes in series electrical current flow through the solution and the rock matrix. Although this model gave good agreement with experiment, it was not developed further. The difficulty lay in not being able to define the geometrical factors in equation (1) more precisely. As we shall see, the importance of the model proposed later lies in its success in overcoming this deficiency.

The Waxman and Smits (W-S) model

Waxman and Smits (1968) proposed a model which has become the most fully developed and widely accepted approach

to the understanding of the electrical conductivity of shaly sand formations. It is based on three assumptions.

(1) The most basic assumption which serves as the groundwork for the model is that the electrical conductance of shaly sand formations can be represented by two resistive elements in parallel. One element represents the resistance associated with the free electrolyte in the pore volume of the rock. The other element represents the resistance associated with the clay whose conductance arises from exchange cations believed to originate from lattice imperfections. Equation (1), therefore, reduces to

$$\sigma_0 = X\sigma_r + Y\sigma_w \quad (2)$$

(2) The second assumption of Waxman and Smits is that the current attributed to the clay exchange ions and the current attributed to the electrolytic ions follow the same path through the formation. It follows that $X = Y$ in equation (2). This means that the two conduction mechanisms must use the same water, i.e., a distinction is not made between, say, bound clay water and pore water. Here, then, one has a single water model where the clay exchange cations and the ions of the electrolyte are uniformly distributed and share the same pore space. This is schematically shown in Figure 1a.

The second assumption is superfluous since it follows from the first assumption. If there are no conduction paths which go through both the rock matrix and the water, then the path through the water must be the same as the path associated with the clay cations, i.e., $X = Y$. Conversely if $X \neq Y$, then the two paths must be spatially separated and a conduction path must also exist which goes through both the rock matrix and the water, i.e., the first term in equation (1) must be taken into account. In any case, if $X = Y$ a conceptual difficulty arises when our current understanding of the electrochemistry of clay minerals, such as smectite, is considered (Van Olphen, 1963). The principal building elements of these minerals are two-

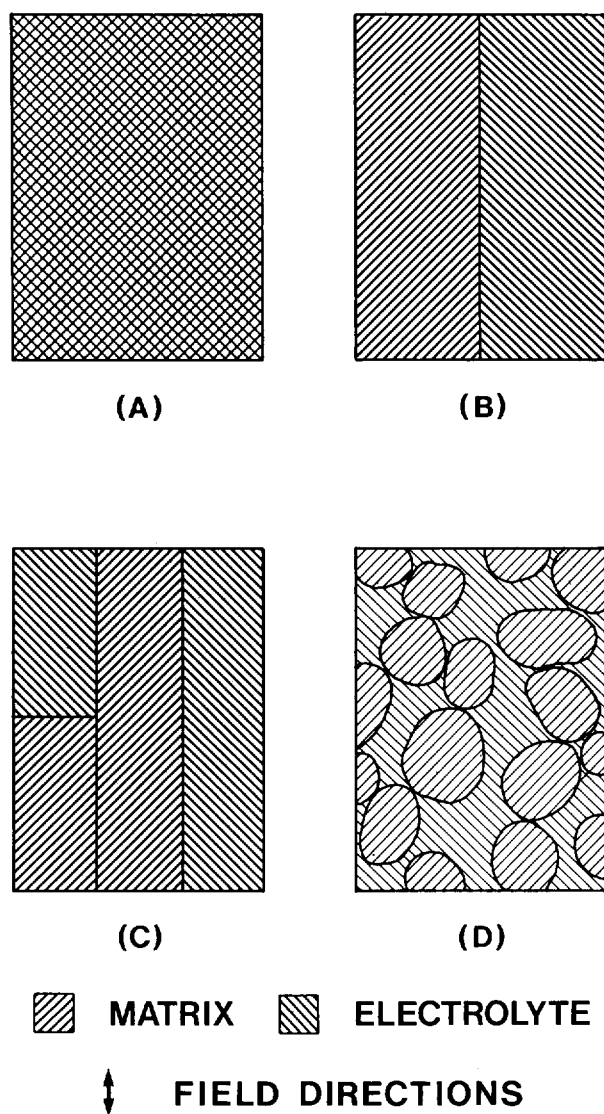


FIG. 1. Geometrical schematics of models discussed in text. The electric field direction is shown by the arrow. (a) Waxman-Smits model assumes matrix and electrolytic conduction paths are identical. (b) Dual water model assumes matrix and electrolytic conduction paths are spatially separated. (c) Three-resistor model is the same as (b) but includes a matrix-electrolyte in series element. (d) Proposed model is a more realistic version of three-resistor model.

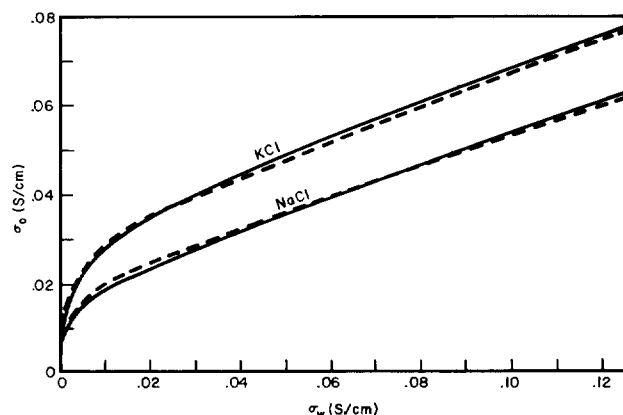


FIG. 2. Conductivity of resin spheres saturated with KCl and NaCl. The dashed curves are fits of equation (1) using the geometrical factors determined by Sauer et al. (1955). Since these fits are in good agreement with experiment, the data points are not shown. The solid curves are equation (8) for $\phi = 0.43$ and $m = 1.3$, with $\sigma_r = 0.047$ S/cm for KCl and $\sigma_r = 0.026$ S/cm for NaCl.

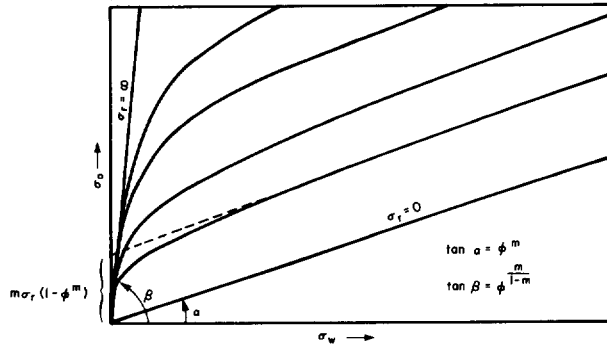


FIG. 3. σ_0 versus σ_w . Family of curves generated from equation (8) with $\phi = 0.2$, $m = 2$, and σ_r from 0 to ∞ .

dimensional (2-D) arrays of silicon-oxygen tetrahedra and 2-D arrays of aluminum- or magnesium-oxygen-hydroxyl octahedra. The combination of an octahedral sheet and one or two tetrahedral sheets is called a unit layer. Most clay minerals consist of a large number of such unit layers which are stacked parallel to each other. This results in an immense surface area. Associated with this multilayered surface is a negative charge which arises from the substitution of aluminum ions for silicon ions within the framework of the clay mineral. To balance this negative charge, exchange cations are distributed near the clay surface and, therefore, cannot be distributed uniformly throughout the electrolyte as Waxman and Smits assumed.

(3) The last assumption of the model is that σ_r is dependent upon σ_w . Now, equation (2) must be written as

$$\sigma_0 = (\sigma_r + \sigma_w)\phi^m, \quad (3)$$

where ϕ = porosity and m = a geometrical factor which characterizes the formation. This result follows if equation (2) is to be consistent with assumption (2) and Archie's empirical law (Archie, 1942), and if, as will be discussed, X and Y are to depend upon geometry only. The model is developed further by assuming that σ_r arises from cation exchange, which when treated as a volume phenomenon gives

$$\sigma_r = \Lambda_+ Q_v, \quad (4)$$

where Λ_+ = equivalent cation conductance in $S\text{-cm}^2/\text{g-equiv}$, and Q_v = concentration of cations in g-equiv/l . It was found that if equations (3) and (4) are to agree with the previously discussed empirical work of Sauer et al, then

$$\Lambda_+ \propto 1 - \beta \exp(-\sigma_w/\gamma),$$

where β and γ are constants. In other words, this functional dependence was introduced to obtain the curvature at small σ_w in Figure 2 and has no theoretical justification. In the three-resistor model and in the proposed model, this curvature follows in a more natural way from the geometry of the system.

In their work, Waxman and Smits assumed that

$$X = Y = 1/F^*,$$

where

$$\lim_{\sigma_r \rightarrow 0} F^* = \phi^{-m}.$$

From their analysis of shaly sands, they found

$$F^* = \phi^{-m}.$$

In general, however, their model allows for F^* to be a function of σ_r . This contradicts the basic assumption that X and Y are geometrical factors. Hence, F^* must be equal to ϕ^{-m} .

The dual water model

An alternative to the Waxman and Smits model was proposed by Clavier et al (1977) in an attempt to take into account indications that two kinds of water are present in shaly sand formations. Their model is based on two major assumptions. (1) The starting point of the dual water model is identical to assumption (1) of the W-S model. (2) The second assumption is that a shaly sand formation behaves as if two kinds of water are present. The first kind is called clay water and is closely associated with clay particles. As in the W-S model, its conductive nature is attributed to clay exchange cations. The second kind of water is called far water, which is the bulk of the water found in the pore space.

The dual water model differs from the W-S model in the manner in which the clay exchange cations are distributed in the formation. In keeping with current concepts of clay electrochemistry, the exchange cations are assumed to be concentrated near the surface of the unit layers which are the building blocks of clay particles. The electrolytic ions, on the other hand, are distributed throughout the pore space. The two conductive mechanisms are, therefore, spatially separated. This is schematically shown in Figure 1b. This separation into two kinds of water which are spatially separated introduces a conceptual difficulty when one visualizes current flow through the formation. Assumption (1) does not allow for current paths which pass through both kinds of water, i.e., in series resistance paths. For a complete description of current flow, such paths must be considered. In other words, the dual water model assumes that $X \neq Y$. It follows, as was previously discussed, that the first term of equation (1) must, therefore, be taken into account. As noted earlier in the three-resistor model, it is the in-series resistance paths which give rise to the curvature in Figure 2. Thus, not only is a conceptual difficulty introduced if one does not consider such paths, but such paths can explain the curvature of Figure 2 in a straightforward way from the geometry of the system alone.

There is another conceptual problem with the dual water model. X and Y are defined to be geometrical factors which should depend upon the geometry of the formation only. However, the dual water model is formulated in such a way that X and Y are also functions of Q_v and σ_w .

PROPOSED MODEL

The basis of the present approach to the problem of electrical conductance in shaly sands lies with the work of Bruggeman (1935) and Hanai (1960, 1961) who derived an equation which relates the electrical properties of a heterogeneous mixture to the properties of the individual components. If one defines the complex conductivity as

$$\sigma^* = \sigma + i\omega\epsilon_0\kappa, \quad (5)$$

the Hanai-Bruggeman equation is given by

$$\frac{\sigma_0^* - \sigma_r^*}{\sigma_w^* - \sigma_r^*} \left(\frac{\sigma_w^*}{\sigma_0^*} \right)^d = 1 - \phi_r, \quad (6)$$

where

- σ_0^* = complex conductivity of the mixture,
- σ_r^* = complex conductivity of dispersed particles,
- σ_w^* = complex conductivity of continuous medium,
- ϕ_r = fractional volume of dispersed particles,
- d = depolarization factor of dispersed particles,
- ω = angular frequency,
- σ = conductivity,
- ϵ_0 = permittivity of free space,
- κ = relative dielectric constant, and
- $i = \sqrt{-1}$.

The original work of Bruggeman and Hanai considered spheres dispersed in a continuous medium, in which case $d = 1/3$. More generally, d depends upon the shape of the particles and can have values between 0 and 1 (Stratton, 1941; Von Hippel, 1954; Sen et al, 1981; Sen, 1981; Mendelson and Cohen, 1982). The value of 0 corresponds to infinitely long cylindrical rods (needles) whose axes are parallel to the electrical field. The value of 1 corresponds to infinitely short cylindrical rods (plates) whose axes are parallel to the electrical field.

It is important to note that a particular value of d does not specify a unique geometrical configuration. For example, the right combination of infinitely long rods parallel to the electric field ($d = 0$) and perpendicular to the electric field ($d = 1/2$) will give a value identical to that of dispersed spheres ($d = 1/3$). It is

apparent, then, that a lattice-like structure can give rise to d values in the same range as dispersed particles. This means that the definitions of σ_r^* and ϕ_r in equation (6) should be broadened to include an immersed network of interconnected particles, as well as dispersed individual particles. This should be kept in mind later when a geometrical factor, defined as $m = 1/(1 - d)$, is empirically determined for core samples saturated with saline water. m is often called the cementation factor.

Equation (6) can be written as

$$\sigma_0^* = \sigma_w^* \phi^m \left(\frac{1 - \sigma_r^*/\sigma_w^*}{1 - \sigma_r^*/\sigma_0^*} \right)^m, \quad (7)$$

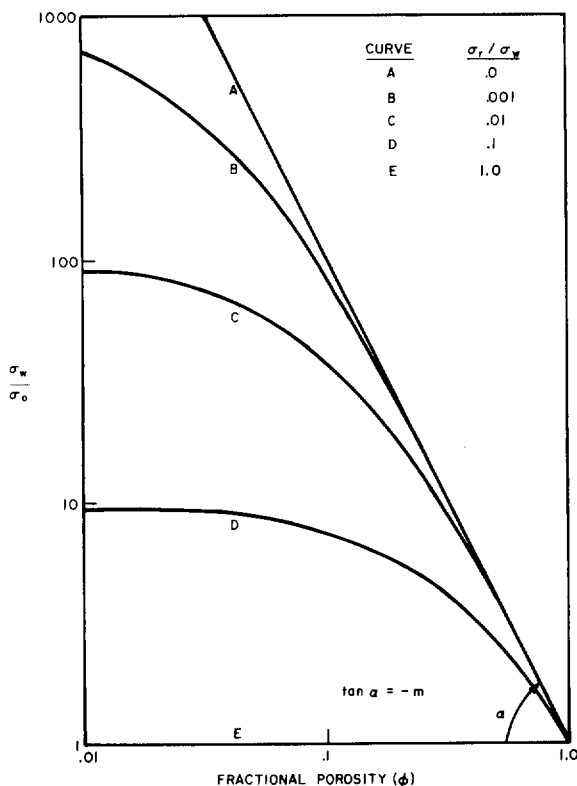
where $\phi = 1 - \phi_r$. Equation (7) is the basic and most general equation of the proposed model. The full implications of this equation are beyond the scope of this paper. In the remainder of this paper discussion will be limited to the low-frequency case with application to shaly sand formations.

At low frequencies equation (7) reduces to

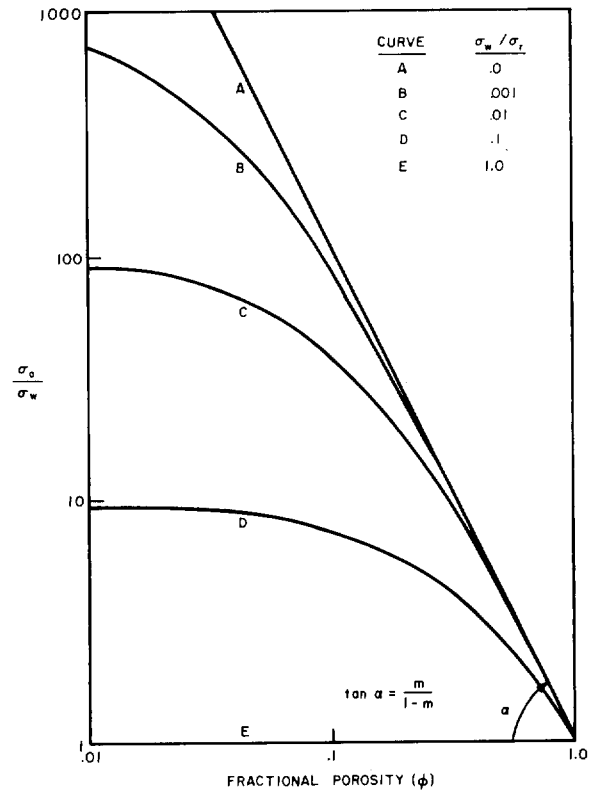
$$\sigma_0 = \sigma_w \phi^m \left(\frac{1 - \sigma_r/\sigma_w}{1 - \sigma_r/\sigma_0} \right)^m. \quad (8a)$$

Equation (8a) can also be written as

$$\sigma_0 = \sigma_w \phi^{m/(1-m)} \left(\frac{1 - \sigma_w/\sigma_r}{1 - \sigma_0/\sigma_r} \right)^{m/(1-m)}. \quad (8b)$$



(a)



(b)

FIG. 4. (a) σ_w/σ_0 versus ϕ . Family of curves generated from equation (13a) for the values of σ_r/σ_w specified and $m = 2$. (b) σ_0/σ_w versus ϕ . Family of curves generated from equation (13b) for the values of σ_w/σ_r specified and $m = 2$.

A σ_0 versus σ_w plot of equation (8) for different σ_r values is shown in Figure 3. The family of curves results from values of σ_r between 0 and ∞ . The limiting cases shown are

$$\lim_{\sigma_r \rightarrow 0} \frac{\sigma_0}{\sigma_w} = \phi^m \quad (9a)$$

and

$$\lim_{\sigma_r \rightarrow \infty} \frac{\sigma_0}{\sigma_w} = \phi^{m/(1-m)} \quad (9b)$$

For the purpose of studying shaly sand formations, it is useful to consider in some detail the case where the water conductivity is greater than the matrix conductivity, i.e., $\sigma_w > \sigma_0 > \sigma_r$. This is the case commonly found in formations as is shown later. Using the binomial expansion, equation (8a) can be written

$$\sigma_0 = \sigma_w \phi^m \frac{1 - m \frac{\sigma_r}{\sigma_w} + \frac{m(m-1)}{2!} \left(\frac{\sigma_r}{\sigma_w}\right)^2 - \dots}{1 - m \frac{\sigma_r}{\sigma_0} + \frac{m(m-1)}{2!} \left(\frac{\sigma_r}{\sigma_0}\right)^2 - \dots} \quad (10)$$

Retaining first-order terms only gives

$$\sigma_0 \approx m\sigma_r(1 - \phi^m) + \phi^m\sigma_w \quad (11)$$

An approximate criterion for the validity of equation (11) can be obtained from equation (10) and is given by

$$m \frac{\sigma_r}{\sigma_0} \gtrsim \frac{10m(m-1)}{2!} \left(\frac{\sigma_r}{\sigma_0}\right)^2, \quad (12)$$

or

$$\sigma_0 \gtrsim 5(m-1)\sigma_r.$$

In some experiments relevant to this paper, data are plotted on a σ_w/σ_0 versus ϕ graph. In order to compare these experiments with the proposed model, one can write equation (8a) in the form

$$\phi = \frac{\left(\frac{\sigma_w}{\sigma_0}\right)^{-1/m} - \left(\frac{\sigma_r}{\sigma_w}\right)\left(\frac{\sigma_w}{\sigma_0}\right)^{(m-1)/m}}{1 - \sigma_r/\sigma_w} \quad (13a)$$

A σ_w/σ_0 versus ϕ plot of equation (13a) for different values of σ_r/σ_w is shown in Figure 4a for the case $m = 2$. The family of curves results from values of σ_r/σ_w between 0 and 1. The importance of this plot will become evident later.

Of academic interest is a symmetry property of equation (13a) which can also be written as

$$\phi = \frac{\left(\frac{\sigma_0}{\sigma_w}\right)^{(1-m)/m} - \left(\frac{\sigma_w}{\sigma_r}\right)\left(\frac{\sigma_0}{\sigma_w}\right)^{1/m}}{1 - \sigma_w/\sigma_r} \quad (13b)$$

A σ_0/σ_w versus ϕ plot of equation (13b) for different values of σ_w/σ_r is shown in Figure 4b for the case $m = 2$. The family of curves results from values of σ_w/σ_r between 0 and 1. The apparent symmetry of Figures 4a and 4b is only true for the case $m = 2$, i.e., for $m = 2$ equation (8) is symmetric under the inverse operations

$$\frac{\sigma_w}{\sigma_0} \rightarrow \frac{\sigma_0}{\sigma_w}$$

and

$$\frac{\sigma_r}{\sigma_w} \rightarrow \frac{\sigma_w}{\sigma_r}.$$

This symmetry is broken for any other value of m . However, symmetry is retained for any value of m if the operation $m \rightarrow m/(m-1)$ is carried out along with the inverse operations.

SHALY SAND FORMATIONS

Earlier the Hanai-Bruggeman equation was reinterpreted and expanded to any two component mixture where one component can be considered to be immersed in the other. This includes the case of a conductive, lattice-like structure immersed in a fluid. Therefore, it is a more geometrically realistic representation of a water-saturated rock formation than any of the models discussed above (Figure 1d).

In equation (8) it is implicitly assumed that σ_r and σ_w are independent variables. If this equation is applied to a shaly sand formation, the implication is that two distinct kinds of water are present. σ_r is then the conductivity resulting, at least in part, from exchange cations in water bound between the unit layers of the clay particles. σ_w is the conductivity arising from the electrolytic ions of the pore water. That is, equation (8) with σ_r and σ_w independent presumes assumption (2) of the dual water model. However, in contrast to the dual water model, the present formulation properly takes into account the spatial separation of these two kinds of water by allowing for current paths through both. The three-resistor model, which did take these series electrical paths into account, was not sufficiently formulated to be of practical value.

In this section experimental results for shaly sand formations are compared with the predictions of equation (8). In spite of the conceptual difficulties discussed earlier, the predictions of the W-S and the dual water models will also be compared with this empirical work.

de Witte (1957) concluded from empirical considerations that the conductive behavior of water-bearing shaly sands could adequately be represented by

$$\sigma_0 = A + B\sigma_w,$$

where A and B are formation parameters independent of σ_w . From equation (11) de Witte's parameters are given by

$$A = m\sigma_r(1 - \phi)^m$$

and

$$B = \phi^m.$$

Not only does this result demonstrate agreement of the proposed model with experiment, but it also enhances the usefulness of de Witte's work. In particular, de Witte has argued on empirical grounds that the water saturation of oil-bearing sands can be determined from the relationship

$$B\sigma_w S_w^2 + A S_w = \sigma_t, \quad (14)$$

where S_w = water saturation and σ_t = measured conductivity of a formation. Since the three-component mixture problem has not yet been solved theoretically, one must still resort to an empirical approach for determining S_w . However, it must be emphasized that although equation (14) was empirically derived, the expressions for A and B are theoretically justified.

The work of de Witte demonstrates that the model agrees with experiment when the brine conductivity is high compared to the rock conductivity, i.e., when equation (12) is satisfied. In

the Sauer et al experiment, which was discussed earlier, a wider range of brine conductivities was used and is a more rigorous test of the model. The conductivity of the resin spheres is believed to be the result of exchangeable ions within the water-filled gel (Spiegler and Coryell, 1951) and in this respect is analogous to clay particles.

The three-resistor model, although it is in agreement with the resin sphere data and suggests that geometry is responsible for the curvature in Figure 2, is not sufficiently developed for a more penetrating interpretation of experimental results. For example, it is unable satisfactorily to explain the dependence on electrolyte type. The proposed model, on the other hand, provides a ready explanation. In Figure 2 approximate fits to equation (8) are compared with fits to equation (1) of the three-resistor model for KCl and NaCl data. It is important to note that in the equation (8) fits the two electrolytes have the same geometrical factors (ϕ and m) but different values for σ_r . The larger σ_r value for the KCl electrolyte can be attributed to the larger ion conductance of K^+ as compared with that of Na^+ . This follows from the procedure used in performing the experiment. Since the resin spheres were thoroughly flushed with the electrolyte of interest before the conductivity measurements were made, K^+ was the exchange ion for the KCl elec-

trolyte and Na^+ was the exchange ion for the NaCl electrolyte.

It should be mentioned that the W-S and dual water models can also explain the shape of the curves in Figure 2, although the latter model appears to breakdown at low values of σ_w . However, a careful comparison of the models in terms of the shape of the curves in Figure 2 is not made here. As will now be shown, a consideration of the σ_w/σ_0 versus ϕ is a more decisive comparison.

Archie (1942) originally proposed the relationship

$$\frac{\sigma_w}{\sigma_0} = a\phi^{-m}, \quad (15)$$

with $a = 1$. Although equation (15) with $a = 1$ is found to hold for "clean" sands, many core measurements require $a \neq 1$ to obtain satisfactory fits (Carothers, 1968; Porter and Carothers, 1970; Gomez-Rivera, 1976). Note that when $a \neq 1$, an inconsistency occurs in the limit

$$\lim_{\phi \rightarrow 1} \sigma_0 = \sigma_w/a.$$

Nevertheless, it has become the accepted practice to analyze core samples by fitting conductivity measurements to equation (15) where a and m are determined. On a log-log plot of σ_w/σ_0

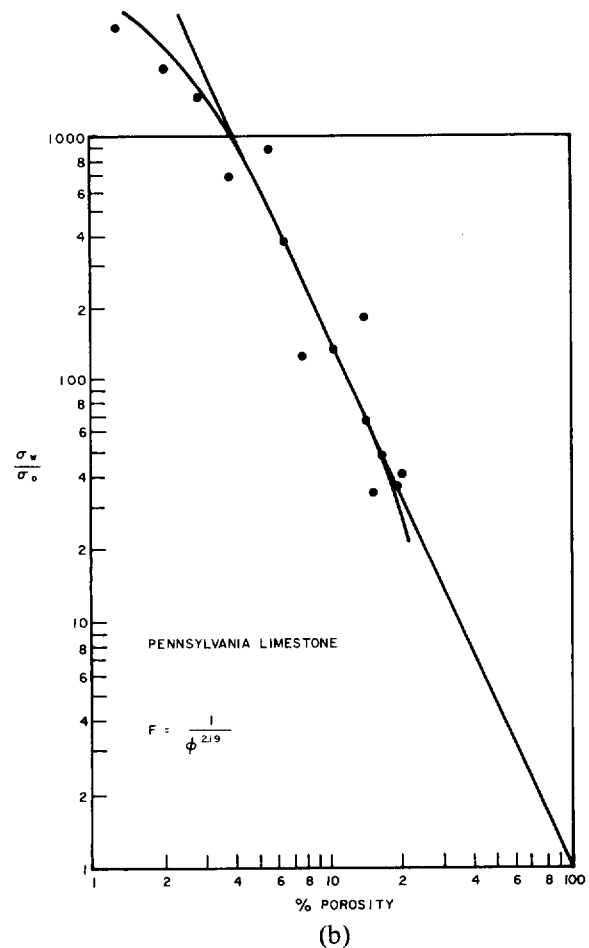
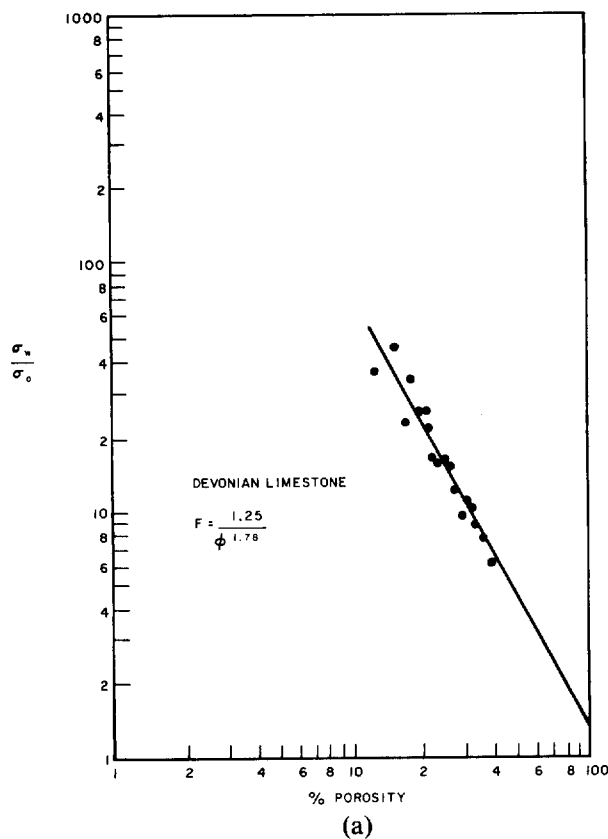


FIG. 5. σ_w/σ_0 versus ϕ . Data and curves are from Carothers (1968).

Table 1. m and σ_r values of cores.
(The first three columns are from Waxman and Smits, 1968)

Sample no.	ϕ (%)	Q_v (equiv/liter)	σ_r (mS/cm)	m	Sample no.	ϕ (%)	Q_v (equiv/liter)	σ_r (mS/cm)	m
25	20.3	.076	.05	1.74	12	11.6	.281	.07	2.00
31	21.8	.081	.09	1.90	13	18.6	.100	.07	1.92
34	21.9	.060	.14	1.87	14	17.4	.206	.13	1.97
38	23.2	.018	.04	1.57	15	16.3	.185	.19	1.97
16	11.8	.008	.01	1.86	1	5.4	.433	.16	2.42
21	13.9	.247	.14	1.75	2	26.1	.191	.25	2.06
25	15.7	.225	.14	1.85	3	25.9	.135	.18	1.98
39	21.2	.033	.12	1.67	4	23.9	.453	.56	1.98
48	22.4	.088	.21	1.79	5	24.8	.200	.31	1.92
5	9.5	.578	.10	1.78	6	25.7	.175	.18	1.97
44	16.5	.232	.08	1.94	1	15.0	2.494	.92	2.50
47	17.5	.215	.07	1.90	2	11.9	2.057	.93	1.71
51	18.6	.073	.03	1.80	3	8.7	1.578	1.09	3.54
52	18.7	.239	.09	1.80	4	25.5	1.436	1.06	2.55
19	20.3	.293	.08	2.19	5	25.3	1.440	1.23	2.63
28	20.1	.327	.11	1.99	6	12.2	1.669	1.10	12.71
36	23.1	.075	.06	1.92	7	27.2	1.332	1.13	2.53
4	16.2	.347	.08	2.12	8	29.0	.689	.70	2.39
13	23.1	.225	.13	2.02	9	26.7	1.074	.85	2.37
17	23.9	.249	.15	2.04	10	26.2	1.228	.91	2.44
18	24.1	.171	.08	1.91	11	28.4	.857	.82	2.29
23	24.5	.195	.11	1.96	12	27.7	1.085	.88	2.40
24	24.5	.146	.12	1.84	13	25.3	1.530	1.07	2.59
27	24.8	.188	.17	1.86	14	23.2	1.541	1.18	2.42
16	23.9	.699	.61	1.99	15	29.8	.776	.85	2.34
22	25.6	.887	.99	2.23	16	12.8	1.587	.91	2.11
27	26.8	.767	.73	2.19	17	27.5	.898	.78	2.34
33	27.4	.798	.71	2.10	18	31.4	.639	.58	2.33
38	28.2	.750	.72	2.19	19	29.0	.922	.81	2.35
42	28.4	.617	.63	2.11	20	32.2	.860	.72	2.61
43	28.6	.645	.68	2.18	21	30.2	.887	.80	2.54
45	29.3	.605	.66	2.15	22	29.0	1.066	.88	2.55
47	30.8	.502	.67	2.17	23	27.2	1.316	1.11	2.51
64	26.5	.148	.09	1.84	1	23.9	.017	.04	1.75
21	17.5	.045	.003	1.95	2	21.2	.052	.06	1.71
1	21.7	.093	.09	1.88	3	23.1	.052	.07	1.65
2	16.6	.302	.16	2.00	4	8.0	.26	.03	1.53
3	24.4	.083	.13	1.85	5	15.4	.20	.13	1.44
4	20.2	.036	.08	1.76	6	21.5	.095	.08	1.90
5	24.2	.051	.09	1.80	7	17.1	.053	.07	1.86
6	21.2	.104	.13	1.82	8	17.1	.053	.06	1.83
7	20.5	.097	.14	1.81	9	19.9	.085	.10	1.79
8	20.8	.069	.17	1.83	10	12.5	.253	.11	7.59
9	20.4	.076	.09	1.79	11	12.5	.253	.01	2.47
10	24.5	.049	.06	1.80	12	11.0	.28	.13	1.71
11	14.5	.454	.20	2.03	13	11.0	.28	.15	1.72
1	20.2	.085	.13	1.86	14	11.0	.28	.21	1.57
2	21.1	.102	.15	1.87	15	9.2	.41	.11	2.24
3	25.8	.050	.08	1.79	16	10.3	.67	.25	1.81
4	19.3	.112	.08	1.95	17	14.0	.33	.36	1.94
5	20.7	.062	.13	1.86	18	25.9	.59	.67	2.16
6	20.0	.067	.12	1.88	19	25.9	.59	.49	2.09
7	18.9	.065	.18	1.86	20	25.9	.59	.54	1.90
8	18.2	.123	.16	1.96	21	23.8	.29	.33	1.87
9	16.1	.158	.09	1.92	22	22.5	.72	.60	2.44
10	14.9	.298	.14	1.89	23	24.2	1.04	.95	2.35
11	15.9	.254	.17	1.97	24	21.6	.81	.66	2.54
					25	18.7	1.27	.79	2.48
					26	22.9	1.47	1.19	2.67
					27	20.9	1.48	1.12	2.53

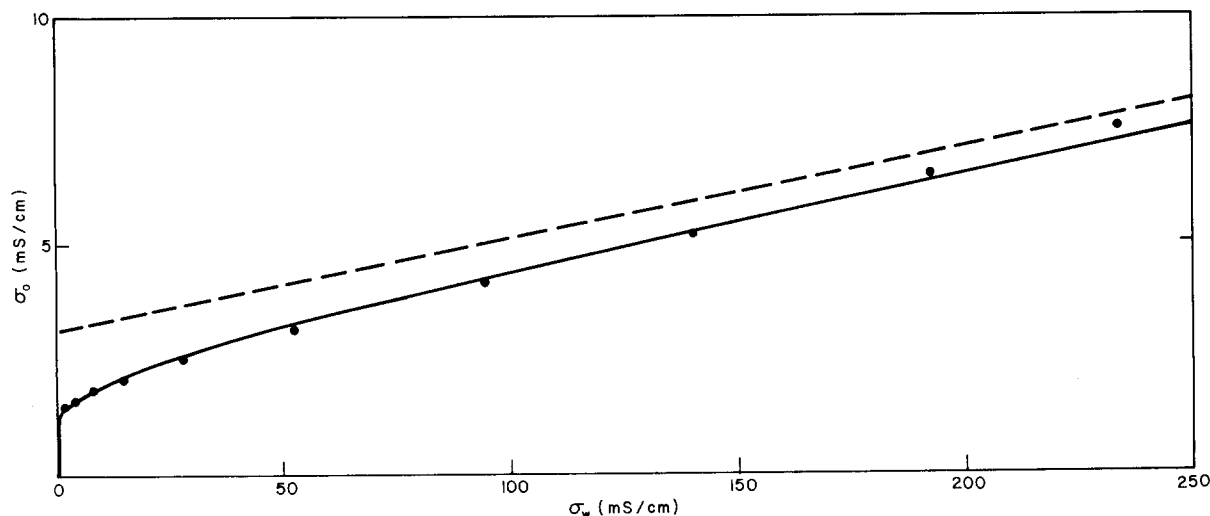


FIG. 6. σ_0 versus σ_w . Data points are from core no. 26 of Table 7 in Waxman and Smits paper. The solid line is equation (8) with $m = 2.67$ and $\sigma_r = 1.19$ mS/cm. The dashed line is the asymptotic limit, i.e., equation (11).

versus ϕ , equation (15) gives a straight line. Two examples taken from Carothers (1968) are shown in Figure 5. These measurements are typical of core data in that $a \geq 1$. Note that for low porosities there appears to be a deviation of the data from equation (15), as is emphasized by the author's curve in Figure 5b. In those cases where core measurements give $a < 1$, a fit with $a = 1$ is within experimental error. For example, a careful analysis of the work of Winsauer et al (1952), reveals that their measurements give a best fit with $a = 0.62$ and $m = 2.15$. Equation (15) with these values is known as the "Humble formula." However, an equally good fit is found with $a = 1$, i.e., the laboratory measurements do not justify $a < 1$.

These effects can be understood in terms of Figure 4a. Since formation porosity values are usually in the 10–40 percent range, the curvature of the family of curves with $\sigma_r \neq 0$ will not be evident in most cases, an example of which is shown in Figure 5a. On the other hand, data which follow these curves but are fit to equation (15) will give values of $a \geq 1$.

In marked contrast, the W-S and dual water models predict $a \geq 1$. This will be shown for the W-S model, although a similar argument can be given for the dual water model. Equation (3) can be written as

$$\frac{\sigma_w}{\sigma_0} = \left(\frac{\sigma_w}{\sigma_r + \sigma_w} \right) \phi^{-m}.$$

A comparison of this equation with equation (15) gives

$$a = \frac{\sigma_w}{\sigma_r + \sigma_w} \leq 1.$$

It has become common practice to classify formations according to the a and m values of equation (15). As seen above, a is an ad hoc parameter with no theoretical basis. Equation (8) suggests a more meaningful classification scheme based on a specification of m and σ_r . Such a program was carried out using conductivity measurements on 117 core samples which were published by Waxman and Smits (1968). A number of conduc-

tivity measurements were made on each core sample, where each measurement was made with the sample being saturated with water of a different salinity. From the published measurements of each core sample, a least-squares fitting program (Bevington, 1969) was applied to equation (8) to determine the values of m and σ_r . These results, as well as independently determined experimental values of ϕ and Q_v , taken from the Waxman and Smits paper, are listed in Table 1.

It is important to note that the exact equation [equation (8)] was used to determine the formation parameters. Since the more conductive matrix cores in Table 1 do not satisfy equation (12), the asymptotic limit given in equation (11) is not justified. An example is shown in Figure 6 with data from a core with a relatively high matrix conductivity. As the curves indicate, the asymptotic limit has not been reached. Furthermore, if equation (11) is used to fit to the data points which appear to lie on a straight line, one obtains values for m and σ_r which are less than their true values.

In Table 2 the average values of σ_r and m are listed for those cases (Tables 1 and 3 of Waxman and Smits, 1968) where core samples were qualitatively classified by the authors. It is clear from this analysis that these categories are only crudely descriptive of the formation. A determination of m and σ_r , on the other hand, provides a more precise method of classification.

The σ_r and m values from Table 1 are cross-plotted in Figure 7. The suggested correlation is probably related to the clay content which can affect both the conductivity of the matrix and its geometrical structure. The correlation is more apparent in Table 3 where the average values of m and σ_r have been determined from core samples of known smectite content. This is an example which illustrates how the formation constitution is reflected in the values of m and σ_r .

Since at least part of the shaly sand matrix conductivity is believed to be due to exchange cations one might expect a correlation between Q_v and σ_r . In Figure 8 a cross plot of these variables for core samples listed in Table 7 of the W-S paper is

Table 2. Average m and σ_r values of cores.

Formation type ¹	Age	Average m^2	Average σ_r (mS/cm)	Average R_r (Ω -m)	Number of samples	Table ⁴
Clean Sandstone Weeks Island, LA	Miocene	$1.80 \pm .08$	$.08 \pm .04$	125	4	1
Clean Sandstone Weeks Island, LA	Miocene	$1.85 \pm .08$	$.12 \pm .04$	83	11	3
Clean Sandstone Mithcell Creek, TX Quitman, TX	Cretaceous	$1.78 \pm .07$	$.12 \pm .06$	83	5	1
Shaly Sandstone Sheridan, TX	Eocene	$1.90 \pm .14$	$.08 \pm .02$	125	6	1
Shaly Sandstone Sheridan, TX	Eocene	$1.91 \pm .06$	$.13 \pm .04$	77	15	3
Shaly Sandstone Seeligson, TX	Oligocene	$1.96 \pm .08$	$.11 \pm .03$	91	9	1
Shaly Sandstone Seeligson, TX	Oligocene	$2.06 \pm .17$	$.27 \pm .14$	37	6	3
Very Shaly Sandstone Big Foot, TX	Cretaceous	$2.15 \pm .07$	$.71 \pm .11$	14	9	1
Very Shaly Sandstone Big Foot, TX	Cretaceous	$2.46 \pm .31$	$.92 \pm .16$	11	22 ³	3
Limestone Chapel Hill, TX	Cretaceous	1.84	.09	111	1	1
Limestone Magnolia, AR	Jurassic	1.95	.003	3333	1	1

¹As quoted by the authors
²Error quoted is one standard deviation
³One sample not included: $m = 12.71$, $\sigma_r = 1.10$ mS/cm
⁴Table number in Waxman and Smits paper (1968).

shown. If matrix conduction is depicted as a volume cation exchange phenomenon, one can use equation (4), but where Λ_+ is now independent of σ_w . For the data shown in Figure 8, one obtains $\Lambda_+ \sim 1$ S-cm²/g-equiv. For comparison $\Lambda_+ = 50$ S-cm²/g-equiv for Na⁺ cations in an electrolyte at infinite dilution, i.e., the limiting equivalent conductance as the concentration goes to zero. One might attribute this reduced exchange cation mobility to the close proximity of charges associated with the clay structure. A knowledge of Λ_+ values, therefore, may be valuable in furthering an understanding of clay structure. Finally, the broad scatter of the data suggests that the matrix conductivity may depend upon additional factors which can only be determined through a more complete understanding of the electro-chemistry of shaly sand formations.

Table 3. Effect of smectite (montmorillonite) on formation parameters.

Smectite (%)	Average σ_r (mS/cm)	Average m	No. of samples
0	$.12 \pm .09$	$1.81 \pm .25$	16
20	$.57 \pm .08$	$2.05 \pm .11$	3
80	$.64 \pm .22$	$2.30 \pm .26$	4
100	$1.03 \pm .17$	$2.56 \pm .08$	3

SUMMARY AND CONCLUSIONS

A model of a rock formation is proposed which is based on the work of Bruggeman and Hanai. By broadening the interpretation of their equation to include a conducting lattice-like matrix immersed in a fluid, a realistic mathematical representation of a rock formation is obtained. In its most general form, both the dielectric constant and the conductivity are taken into account. In the low-frequency limit where displacement current is unimportant, a comparison with the Waxman-Smits and dual water models shows that only the proposed model is free of conceptual difficulties and in agreement with experimental results. This comparison is summarized in Table 4. Column 1 recapitulates the discussion on the conceptual problems of the three models. The number of variables in the equation corresponding to each model is listed in column 2. The fact that the proposed model has the fewest number of variables is not only aesthetic, but it also means that it can be more readily used. As summarized in column 3, only the proposed model agrees with those experiments where the dependence on porosity is studied. Finally, as column 4 indicates, only the proposed model has the potential for expansion to include dielectric constant phenomena which must be incorporated in any fully developed model.

Table 4. Comparison of models.

Model	(1) Conceptual Problems	(2) No. of Variables	(3) Archie's a	(4) Dielectric Applicability
Waxman-Smiths	Yes	8	≤ 1	No
Dual water	Yes	9	≤ 1	No
Proposed	No	5	≥ 1	Yes
Required	No	Minimum	≥ 1	Desirable

The independence of σ_r and σ_w has important implications for understanding shaly sand formations. When taken together with knowledge of clay structure, it supports one of the basic tenets of the dual water model—the presence of two kinds of water. However, in the proposed model the assumption of two resistive elements in parallel is replaced with a more realistic geometrical picture which properly accounts for conducting paths which pass through both kinds of water in series.

The proposed model is applicable to any formation where a distinct conductive property can be ascribed to the matrix. It is fortuitous that shaly sand formations are included in this general, yet simple, representation.

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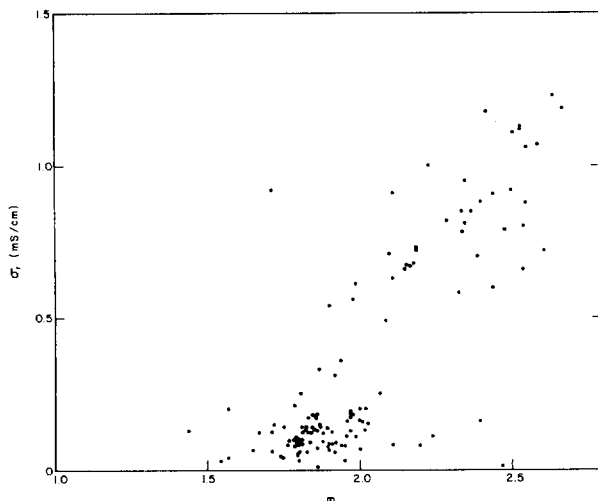


FIG. 7. Matrix conductivity versus cementation factor. Data plotted are from Table 1.

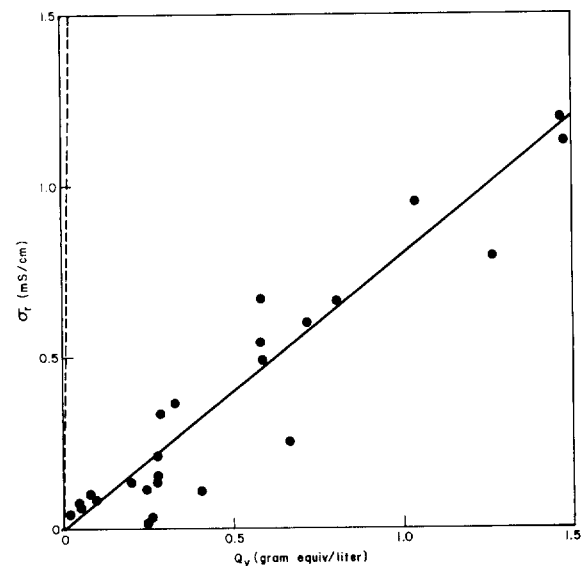


FIG. 8. σ_r versus Q_v . Data plotted are the last group of Table 1. The solid line is equation (4) with $\Lambda_+ = 0.8 \text{ S-cm}^2/\text{g-equiv.}$ The dashed line is equation (4) with $\Lambda_+ = 50 \text{ S-cm}^2/\text{g-equiv.}$

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